

Dulitha Prasanna Kulathunga

Postdoctoral Research Associate

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Summary

- Computational chemist working on data-efficient discovery. I combine active machine learning, Bayesian optimization, and molecular simulation to explore chemical and protein sequence space with as few experiments as possible.

Education

Iowa State University & Ames National Laboratory , Physical Chemistry	Ames, IA
• Thesis advisors: Prof. Theresa L. Windus and Prof. Davit Potoyan. • Dissertation: <i>Exploration without Exhaustion</i>, defended April 10, 2026.	2021 – Apr 2026
University of Colombo , Computational Chemistry	Colombo, Sri Lanka
• Completed 2020. Advisor: Prof. Samantha Weerasinghe. • CUFSAA Gold Medal for Best Performance in Computational Chemistry (2019).	
University of Colombo School of Computing (UCSC) , Information Technology	Colombo, Sri Lanka
• Completed 2019.	

Experience

University of North Carolina at Chapel Hill , Postdoctoral Research Associate, Miao Lab	Department of Pharmacology, Chapel Hill, NC
• Gaussian accelerated molecular dynamics (GaMD) enhanced-sampling simulations of RNA and RNA–ligand systems. • Developing AI-driven models for RNA-targeted drug discovery and virtual screening.	2026 – present 1 year
Iowa State University & Ames National Laboratory , Graduate Research Assistant	Ames, IA
• AMLRO: built an open-source active learning framework for single- and multi-objective reaction optimization, with a Flask/Streamlit GUI for open-loop use and SHAP-based explainability. Applied it to Paal-Knorr flow synthesis, closed-loop diglycolamide (DGA) flow synthesis, and aldehyde-selective Wacker oxidation in a 7-dimensional parameter space. • CyRxnOpt: co-developed a unified interface for benchmarking reaction optimization algorithms, integrated into the RxnRover automation ecosystem. • ALSEBO: developed a coevolution-informed Bayesian optimization framework over a generative latent sequence landscape for data-efficient protein design; designed novel green fluorescent protein variants validated by MD simulations. • Enzyme thermal adaptation: studied serine proteases and cold-adapted β-glucosidases from psychrophilic to thermophilic organisms using molecular dynamics, bioinformatics, and unsupervised machine learning. • Collaborated with Oak Ridge National Laboratory and the Critical Materials Innovation Hub (CMI) on multi-objective optimization of molten salt electrochemical graphitization.	2021 – Apr 2026 5 years
Iowa State University , Graduate Teaching Assistant	Ames, IA
• General chemistry laboratories and recitations (Chem 163/163L, 167L, 177L).	2021 – 2022 1 year
University of Colombo , Teaching Assistant	Colombo, Sri Lanka
• Computational chemistry laboratory courses for undergraduate and M.Sc. students.	2020 – 2021 1 year

University of Colombo, Undergraduate Research Assistant

Colombo, Sri Lanka

- Compared all-atom (AMBER99SB-ILDN) and united-atom (KBFF) force fields for interactions between β -amyloid and model amino acids with unsaturated side chains, using GROMACS.

2019 – 2020

1 year

Open Source Projects

AMLRO (Active Machine Learning Reaction Optimizer)

Active learning framework for single- and multi-objective optimization of chemical reaction conditions.

- Closed-loop (automated) and open-loop (human-in-the-loop) optimization, with a web GUI.
- Explainable AI with SHAP to rank the influence of reaction variables.
- Released as DOE software: [doi:10.11578/dc.20260205.1](https://doi.org/10.11578/dc.20260205.1).

ALSEBO (Active Learning Sequence Exploration via Bayesian Optimization)

Generative protein design by Bayesian optimization in a learned latent space.

- Variational autoencoder (VAE) latent space for protein sequences.
- Coevolution-informed Bayesian optimization to propose new variants with few measurements.
- Demonstrated on avGFP fluorescence ([bioRxiv 2026](#)).

CyRxnOpt

Common Python interface for reaction optimization algorithms, usable from a CLI or inside self-optimizing platforms.

Awards

Ames National Laboratory Innovation Incentive Award (for CyRxnOpt)

2026

Ames National Laboratory

Klaus Ruedenberg Theoretical Chemistry Award

2025

Department of Chemistry, Iowa State University

Drtina Chemistry Graduate Student Travel Award

2025

Received in 2024 and 2025.

Department of Chemistry, Iowa State University

Edward V. Sayre Chemistry Scholarship

2024

Department of Chemistry, Iowa State University

CUFSAA Gold Medal for Best Performance in Computational Chemistry

2019

University of Colombo

Skills

Programming & Apps

Machine Learning

Molecular Simulation

Computational Chemistry & Cheminformatics